Name _____ _____ Baggett/Fall2026/207/Quiz4

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{f'(x)}{g'(x)}$.

True **False**

Why? That is not the Quotient Rule, so try to break it. Pick an f and a g you can divide and differentiate in your head, but not so simple that both sides collapse to the same thing — take $f(x) = x^2$ and $g(x) = x$. Left side: $f/g = x^2/x = x$, whose derivative is 1. Right side: $f'(x) = 2x$ and $g'(x) = 1$, so $f'/g' = 2x/1 = 2x$. At $x = 3$ the left side is 1 and the right side is 6. They disagree, so the statement is false.

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2. (**2** pts) If $f(x) = \sin x$, then $f^{(19)}(x) = \cos x$.

True **False**

Why? **The derivatives repeat every four:** $\sin x \to \cos x \to -\sin x \to -\cos x$**. Since** $19 = 4(4) + 3$**, the nineteenth is the third in the cycle:** $f^{(19)}(x) = -\cos x$**.**

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3. (**5** pts) Let $f(x) = x\cos x$.

(a) Find $f'(x)$.

(b) Find the equation of the **tangent line** to $y = f(x)$ at $x = \pi$.

(a) $f'(x) = \cos x - x\sin x$ **(b)** $y = -x$

(a)	$f'(x) = (1)\cos x + x\,(-\sin x) = \cos x - x\sin x$	Product Rule
(b)	$f(\pi) = \pi\cos\pi = -\pi$	the point is $(\pi,\ -\pi)$
	$f'(\pi) = \cos\pi - \pi\sin\pi = -1 - 0 = -1$	the slope
	$y - (-\pi) = -1\,(x - \pi) \implies y = -x$	point-slope

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4. (**5** pts) Differentiate $f(x) = \dfrac{\sin x}{x^2+1}$.

$$f'(x) = \frac{\left(x^2+1\right)\cos x - 2x\sin x}{\left(x^2+1\right)^2}$$

$$f'(x) = \frac{\left(x^2+1\right)\left(\frac{d}{dx}\sin x\right) - \sin x\left(\frac{d}{dx}\left(x^2+1\right)\right)}{\left(x^2+1\right)^2} \qquad \text{lo D hi minus hi D lo, over lo squared}$$

$$= \frac{\left(x^2+1\right)\cos x - 2x\sin x}{\left(x^2+1\right)^2}$$

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5. (**3** pts) Differentiate $y = x^3 + e^x\cos x + 2x\sin x$.

$y' = 3x^2 + e^x\cos x - e^x\sin x + 2\sin x + 2x\cos x$

$$\frac{d}{dx}\left(x^3\right) = 3x^2 \qquad \text{Power Rule}$$

$$\frac{d}{dx}\left(e^x\cos x\right) = e^x\cos x + e^x\left(-\sin x\right) = e^x\cos x - e^x\sin x \qquad \text{Product Rule}$$

$$\frac{d}{dx}\left(2x\sin x\right) = 2\sin x + 2x\cos x \qquad \text{Product Rule}$$

$$y' = 3x^2 + e^x\cos x - e^x\sin x + 2\sin x + 2x\cos x$$

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6. (**3** pts) For each function below, name the **rule you would use first**. **Do not differentiate anything.**

(a) $y = x^2\cos x$ _____

(b) $y = \dfrac{x^2}{3}$ _____

(c) $y = \dfrac{\cos x}{x^2}$ _____

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(a) Product Rule. **(b)** Power Rule — the denominator is the *constant* 3, so this is $\frac{1}{3}x^2$ and no Quotient Rule is needed. **(c)** Quotient Rule.